

ch-4

①

① Definition of PDA : CSE Academy screenshot
- collection of $\rightarrow$ tuples/components $(Q, \Sigma, q_0, F, Z_0, \Gamma, \delta)$
why it come if FA exists? $\rightarrow$ bcz FA has limited amount of memory.

PDA = FA + Memory element (stack)

FA accepts RL

PDA accepts CFL

$\Rightarrow$ RL : No counting $\rightarrow$ e.g. $ab, (a+b)^+, a^+b^+$
 $\rightarrow$ simple pattern e.g. string ending with 01

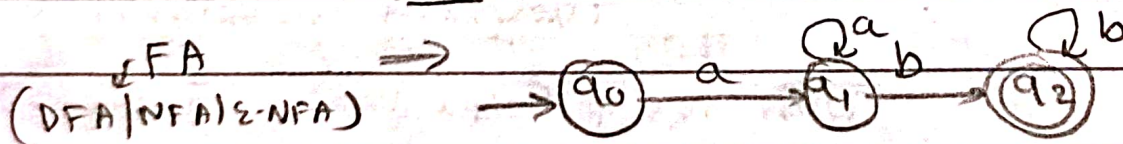
CFL : Need counting $\rightarrow$ e.g. $a^n b^n, 0^n 1^n$ } $\rightarrow$ maximum two times same power should come.

but if more than two counting $\rightarrow$ NOT CFL

e.g. $a^n b^n c^n$

RL $\Rightarrow L = a^n b^m \mid n, m \geq 1$

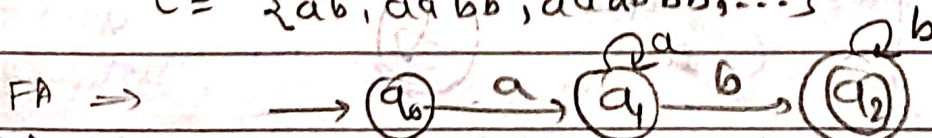
$L = \{ab, abb, aab, aaabb, \dots\}$



here, string which is not in L should not be accepted by FA. e.g. bba

CFL $\Rightarrow L = a^n b^n \mid n > 0$

$L = \{ab, aabb, aaabbb, \dots\}$



$\rightarrow$ so, PDA came in picture

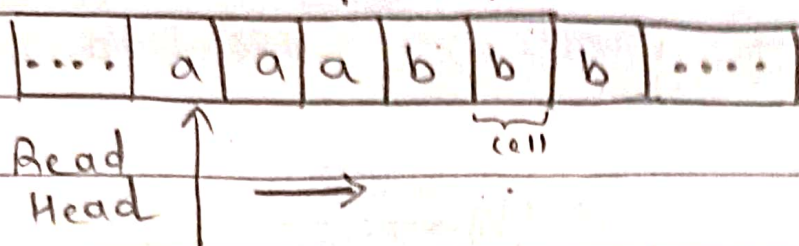
e.g. abb, but abb is accepted by FA of given CFL.

② → CSE academy

* Structure / Model of PDA e.g. $a^n b^n \Rightarrow a a a b b b$

input tape

element



Finite
(control)

- 1) Push
- 2) Pop
- 3) Read

Z_0

stack

4 elements :

- 1) input tape
- 2) Read Head
- 3) Finite control
- 4) stack

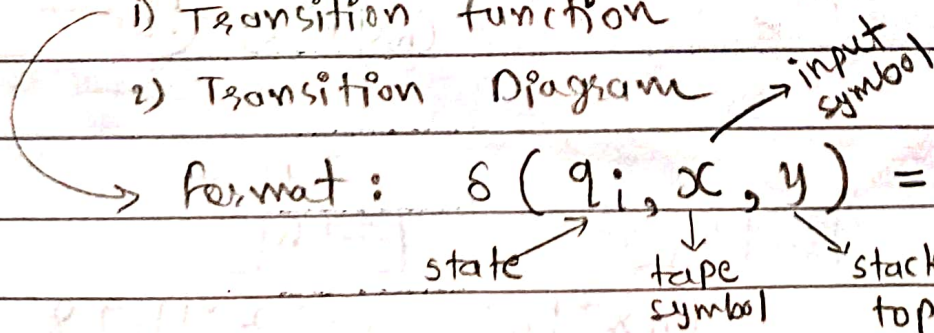
+ initial top of stack is Z_0

before & after operation, stack should be empty.

* Representation of PDA

1) Transition function

2) Transition Diagram



format : $\delta(q_i, x, y) = (q_j, xy)$

state

tape symbol

stack top

state

stack top

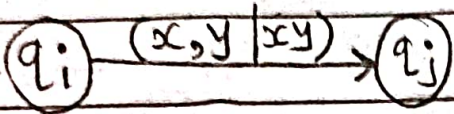
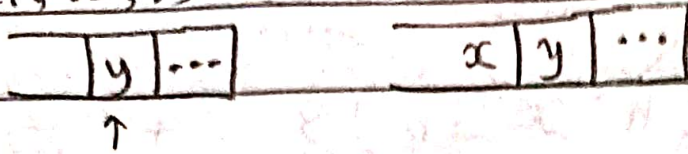
Before

after

3 operations of stack (3) cse academy.

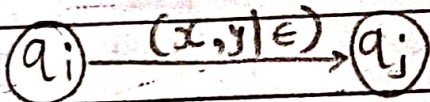
1) For Push :

$$\delta(q_i, x, y) = (q_j, xy)$$



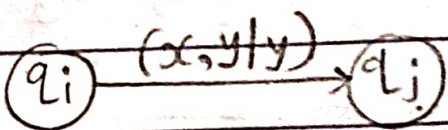
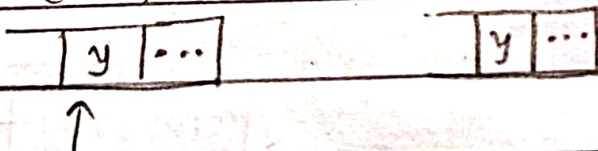
2) For Pop :

$$\delta(q_i, x, y) = (q_j, \epsilon)$$



3) For Read :

$$\delta(q_i, x, y) = (q_j, y)$$



* Design PDA for $L = a^n b^n \mid n \geq 1$

$a^n b^{2n}, n \geq 0 \mid n \geq 2$
 $0^n 1^{2n}, n \geq 0$
 → create smashes.

1) Analysis

↪ CSE academy.
 (H.A)
 (21.03)

$n=1, ab$

$n=2, aabb$

$n=3, aaabbb$

2) Algorithm

1) push 'a' into stack.

2) pop corresponding 'b' as read.

a
a
a
z0

3) Transition function:

1) $\delta(q_0, a, z_0) = (q_0, a z_0) \rightarrow$ Push operation

2) $\delta(q_0, a, a) = (q_0, aa) \rightarrow$ Push

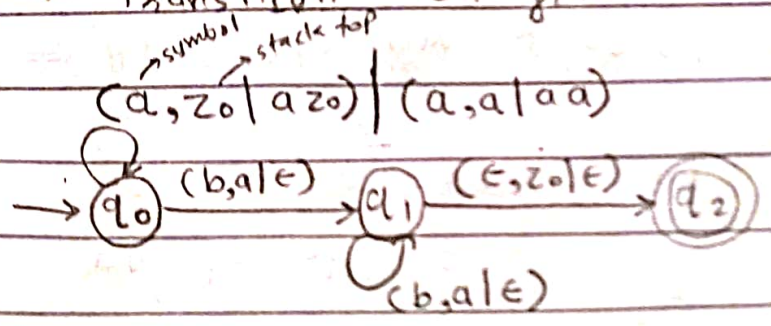
3) $\delta(q_0, b, a) = (q_1, \epsilon) \rightarrow$ Pop a

4) $\delta(q_1, b, a) = (q_1, \epsilon) \rightarrow$ Pop a

5) $\delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$

↪ acceptance by empty stack

4) Transition diagram



ਜਾਂ ਸੁਰ/ input symbol
 'b' ਏਲ/ ਤਦ stack
 top 'a' ਏਲ/
 ਜਾਂ ਸੁਰ/ POP ਕਰਨੀ
 ਚਾਹੀਦੀ.

YT: Knowledge Gate Jay

5) Verification

- | | |
|--------------------------------|---------------------------------|
| $\vdash (q_0, aaabbb, z_0)$ ① | $\vdash (q_1, bb, aa z_0)$ ④ |
| $\vdash (q_0, aabbb, a z_0)$ ② | $\vdash (q_1, b, a z_0)$ ④ |
| $\vdash (q_0, abbb, aa z_0)$ ③ | $\vdash (q_1, \epsilon, z_0)$ ⑤ |
| $\vdash (q_0, bbb, aaa z_0)$ ③ | $\vdash (q_2, \epsilon)$ |

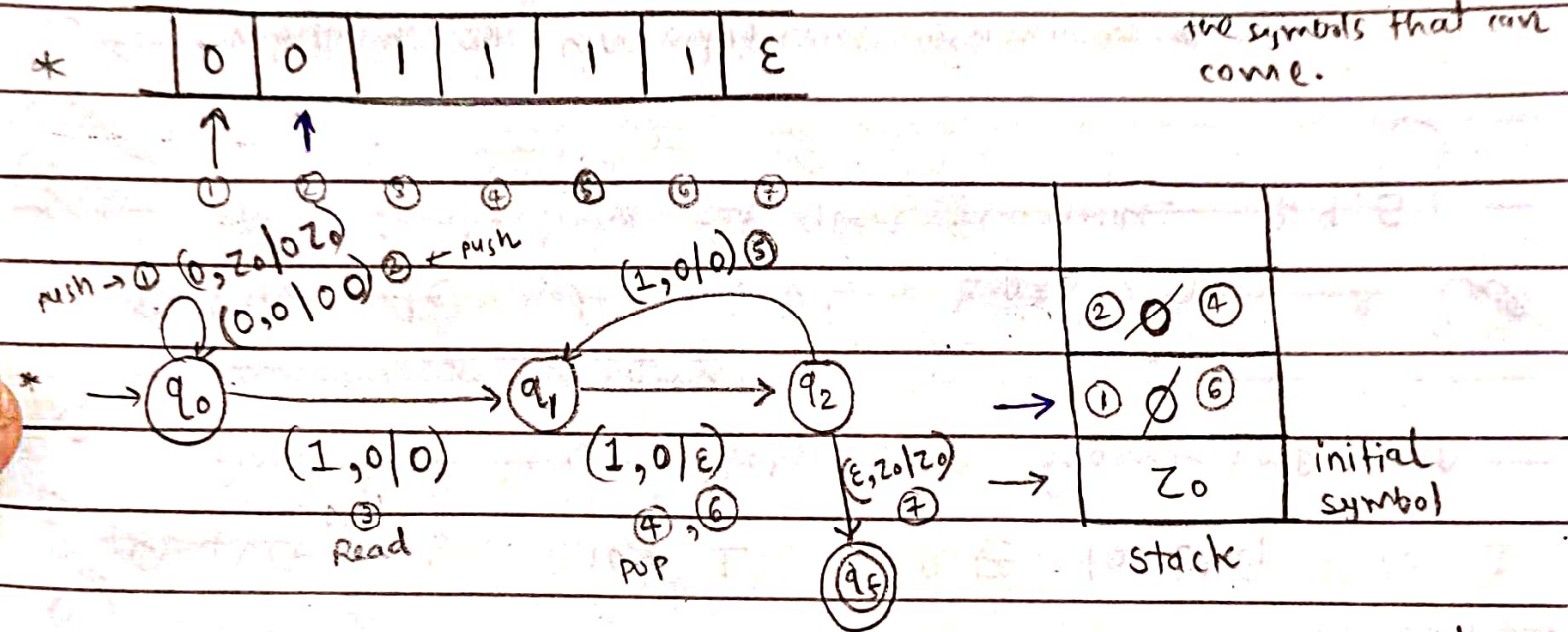
15 $0^n 1^{2n}$, $n > 0$: Gate smashers. } design PDA

$L = \{ 011, 001111, 000111111, \dots \}$

here, we need to pop two 1's, corresponding to one 0

ie. for eg 001111

Γ = in stack, which are the symbols that can come.



* $\Gamma = 0, Z_0$ $\xrightarrow{\text{stack top}}$ already in stack

$\epsilon = 0, 1$

here, $(0, 0)$ means - $\downarrow$ input $\xrightarrow{\text{instack}}$ push 2nd 0 on stack

* Transitions.

$q_0, 0, Z_0 \xrightarrow{q_0} 0Z_0$

$q_0, 0, 0 \xrightarrow{q_0} 00$

$q_0, 1, q_1 \xrightarrow{q_1} 0$

$q_1, 1, 0 \rightarrow q_2, \epsilon$

$q_2, 1, 0 \rightarrow q_1, 0$

$q_2, \epsilon, Z_0 \rightarrow q_f, Z_0$

6 CSE academy.

* Design a PDA for $a^n b^n \mid n \geq 1$

$a^n b^n \mid n \geq 1$
 $a^n b^n \mid n \geq 0$

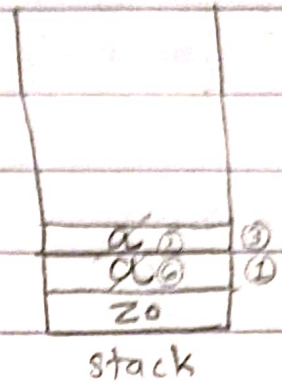
Solution: 1) Analysis

$n=1, a^2 b^1 = aab$

$n=2, aaaa b^2$

$n=3, aaaaaa b^3$

$L = \{aab, aaaa b^2, aaaaaa b^3, \dots\}$

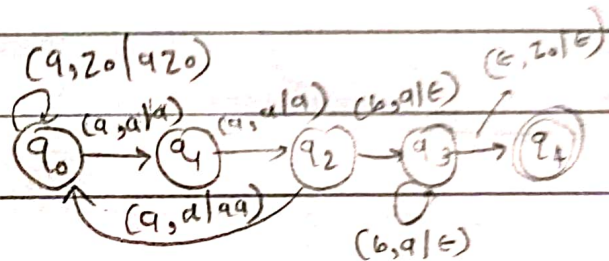


2) Algorithm

select any one string: $a a a a b b$
 Push Read Pop Pop

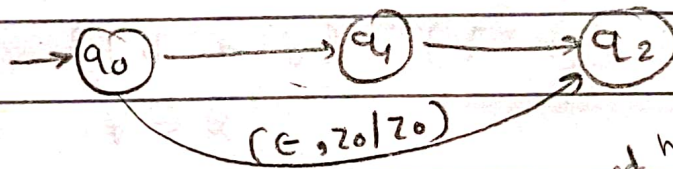
if: $a^n b^n \Rightarrow a a a b$

$a a a a a b b b$
 push R R

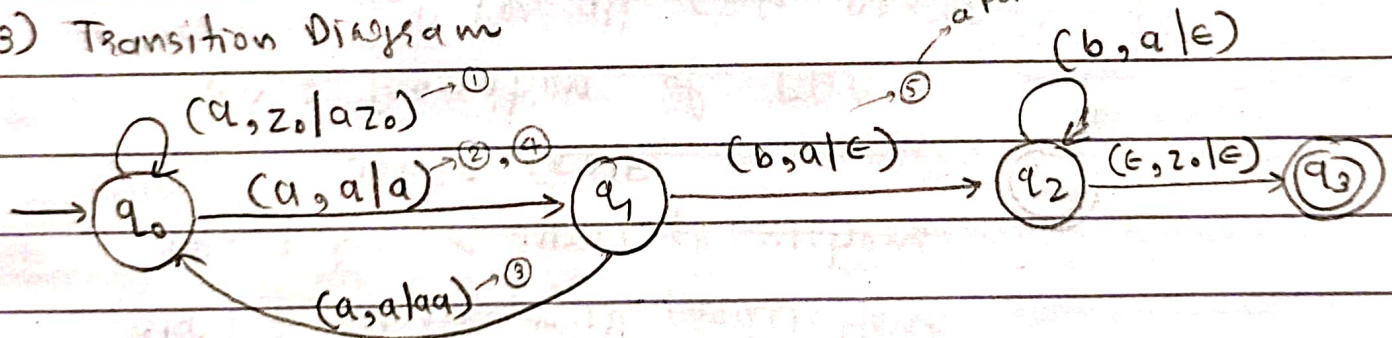


if $a^n b^n \mid n \geq 0$

$n=0 \rightarrow a^0 b^0 = \epsilon$



3) Transition Diagram



4) Transition function

1) $\delta(q_0, a, Z_0) = (q_0, aZ_0)$

5) $\delta(q_2, b, a) = (q_2, \epsilon)$

2) $\delta(q_0, a, a) = (q_1, a)$

6) $\delta(q_2, \epsilon, Z_0) = (q_3, \epsilon)$

3) $\delta(q_1, a, a) = (q_0, aaa)$

4) $\delta(q_1, b, a) = (q_2, \epsilon)$

if symbol stack top

⑦ CFG to PDA conversion

→ Equivalence between CFG and PDA

- Conversion of CFG to PDA
- Conversion of PDA to CFG

(1) Conversion of CFG to PDA

Step 1 : Remove null productions

Step 2 : Remove unit productions

Step 3 : Obtain CNF

Step 4 : Transition of PDA

1) For all CFG productions

$$A \rightarrow \alpha$$

$$\delta(q, \epsilon, A) = (q, \alpha)$$

here, A = (left symbol)

α = right production

2) for all terminals

$$a \in T$$

$$\delta(q, a, a) = (q, \epsilon)$$

CFG to PDA example:

GNF: 1st symbol terminal (single terminal only) & after that any no of non-terminal symbols. one length: terminal
more than one length: termi. Nontermi

$$\textcircled{1} S \rightarrow aSa \mid aSb \mid a$$

as there is no null productions & no unit productions,
we will check whether given CFG is in GNF form or not.

- given CFG is not in GNF

$S \rightarrow aSa$
 $S \rightarrow aSb$ } $\rightarrow$ Not in GNF bcz 1st symbol is terminal, after that non terminal, but 3rd symbol is again terminal symbol. so, both are not in GNF.

so, obtain GNF

$$S \rightarrow aSA \mid aSB \mid a$$

$$A \rightarrow a$$

$$B \rightarrow b$$

$$G = (V, T, P, S)$$

$$V = \{S, A, B\}$$

$$T = \{a, b\}$$

$$P = \{S \rightarrow aSA \mid aSB \mid a$$

$$A \rightarrow a$$

$$B \rightarrow b\}$$

Transition of PDA

(1) For all CFG productions

i) $S \rightarrow aSA$

$$\delta(q, \epsilon, S) = (q, aSA)$$

ii) $S \rightarrow aSB$

$$\delta(q, \epsilon, S) = (q, aSB)$$

iii) $S \rightarrow a$ $\delta(q, \epsilon, S) = (q, a)$

iv) $A \rightarrow a$

$$\delta(q, \epsilon, A) = (q, a)$$

v) $B \rightarrow b$

$$\delta(q, \epsilon, B) = (q, b)$$

(2) For all terminals

(i) for 'a'

$$\delta(q, a, a) = (q, \epsilon)$$

(ii) for 'b'

$$\delta(q, b, b) = (q, \epsilon)$$

* TF of PDA

$$1) \delta(q, \epsilon, s) = (q, aSA)$$

$$2) \delta(q, \epsilon, s) = (q, aSB)$$

$$3) \delta(q, \epsilon, s) = (q, a)$$

$$4) \delta(q, \epsilon, A) = (q, a)$$

$$5) \delta(q, \epsilon, B) = (q, b)$$

$$6) \delta(q, a, a) = (q, \epsilon)$$

$$7) \delta(q, b, b) = (q, \epsilon)$$

$$(2) \text{ CFG} \rightarrow G = (\{S, A\}, \{0, 1\}, P, S)$$

P is defined by:

$$S \rightarrow OS/A$$

$$A \rightarrow 1A0/S/\epsilon$$

} CFG to PDA

1) find nullable variables

A, S

Remove null productions

$$S \rightarrow OS/O/A$$

$$A \rightarrow 1A0/10/S$$

2) find unit productions

A, S on right side

- remove unit productions.

$$(1) S \rightarrow OS/O/1A0/10/S$$

$$A \rightarrow 1A0/10/S$$

$$(2) S \rightarrow OS/1A0/10/O$$

$$A \rightarrow 1A0/S/10$$

$$(3) S \rightarrow OS/1A0/10/O$$

$$A \rightarrow 1A0/OS/1A0/10/O/10$$

3) obtain GNF

$$S \rightarrow OS/1AB/1B/O$$

$$B \rightarrow O$$

$$A \rightarrow 1AB/OS/1AB/1B/O/1B$$

4) Transition of PDA

i) transition for all CFG productions

$$(1) \delta(q, \epsilon, S) = (q, OS)$$

$$(2) \delta(q, \epsilon, S) = (q, 1AB)$$

$$(3) \delta(q, \epsilon, S) = (q, 1B)$$

$$(4) \delta(q, \epsilon, S) = (q, O)$$

$$(5) \delta(q, \epsilon, B) = (q, O)$$

$$(6) \delta(q, \epsilon, A) = (q, 1AB)$$

$$(7) \delta(q, \epsilon, A) = (q, OS)$$

$$(8) \delta(q, \epsilon, A) = (q, 1AB)$$

$$(9) \delta(q, \epsilon, A) = (q, 1B)$$

$$(10) \delta(q, \epsilon, A) = (q, O)$$

$$(11) \delta(q, \epsilon, A) = (q, 1B)$$

ii) transition for all terminals

$$(1) \delta(q, O, O) = (q, \epsilon)$$

$$(2) \delta(q, 1, 1) = (q, \epsilon)$$

PDA $\rightarrow$ CFG - CSE Academy

Q. Obtain CFG

$\rightarrow$ initial stack top

$$M = (\{q_0, q_1\}, \{0, 1\}, \{\epsilon, x\}, \delta, z_0, q_0, \emptyset)$$

δ is defined as

$$1) \delta(q_0, 1, z_0) = (q_0, xz_0)$$

$$2) \delta(q_0, 1, x) = (q_0, xx)$$

$$3) \delta(q_0, 0, x) = (q_1, x)$$

$$4) \delta(q_1, \epsilon, z_0) = (q_0, \epsilon)$$

$$5) \delta(q_1, 1, x) = (q_1, \epsilon)$$

$$6) \delta(q_0, 0, z_0) = (q_0, z_0)$$